

Data Analyst – Internship

Project: 2 – Startup Investment (Shark Tank India Datasets)

After project completion, prepare a 1–2 page report in PDF format, containing:

Introduction

Abstract

Tools Used

Steps Involved in Building the Project

Conclusion

Note: Report must not exceed 2 pages.

```
Python Console
...: import pandas as pd
...: import os
...: import matplotlib.pyplot as plt
>>> ...: from matplotlib.pyplot import xlabel
...: from matplotlib.pyplot import ylabel
...: from matplotlib.pyplot import title
...: from matplotlib.pyplot import show
...: import seaborn as sns

Backend tkagg is interactive backend. Turning interactive mode on.

In [3]: df = pd.read_csv('Shark Tank India Dataset.csv')
In [4]: df_sk_n = pd.read_csv('Shark Tank Data complete 121pitches.csv')
In [5]: df.head()
Out[5]:
  episode_number  pitch_number  ... amount_per_shark equity_per_shark
0              1              1  ...             25.0             5.333333
1              1              2  ...             20.0            25.000000
2              1              3  ...             12.5            15.000000
3              2              4  ...             70.0             2.750000
4              2              5  ...              0.0             0.000000

[5 rows x 28 columns]

In [6]: df_sk_n.head()
Out[6]:
   Sl No.  Start-Up Name  ...  Unnamed: 23  Unnamed: 24
0      1  BluePine Industries  ...         NaN         NaN
1      2      Booz scooters  ...         NaN         NaN
2      3  Heart up my Sleeves  ...         NaN         NaN
3      4      Tagz Foods  ...         NaN         NaN
4      5    Head and Heart  ...         NaN         NaN

[5 rows x 25 columns]

In [8]: df.shape
Out[8]: (117, 28)
In [9]: df_sk_n.shape
Out[9]: (121, 25)

In [10]: |
```

```
In [10]: df.describe()
Out[10]:
```

	episode_number	pitch_number	...	amount_per_shark	equity_per_shark
count	117.000000	117.000000	...	117.000000	117.000000
mean	18.735043	59.000000	...	18.132481	5.583590
std	10.070778	33.919021	...	23.588682	10.803799
min	1.000000	1.000000	...	0.000000	0.000000
25%	10.000000	30.000000	...	0.000000	0.000000
50%	19.000000	59.000000	...	10.000000	1.250000
75%	27.000000	88.000000	...	25.000000	6.000000
max	35.000000	117.000000	...	100.000000	75.000000

[8 rows x 26 columns]

```
In [11]: df_sk_n.describe()
Out[11]:
```

	Sl No.	No. of Pitchers	...	Unnamed: 23	Unnamed: 24
count	121.000000	121.000000	...	0.0	0.0
mean	61.000000	2.181818	...	NaN	NaN
std	35.073732	1.008299	...	NaN	NaN
min	1.000000	1.000000	...	NaN	NaN
25%	31.000000	2.000000	...	NaN	NaN
50%	61.000000	2.000000	...	NaN	NaN
75%	91.000000	3.000000	...	NaN	NaN
max	121.000000	6.000000	...	NaN	NaN

[8 rows x 18 columns]

```

In [12]: df.info()
<class 'pandas.DataFrame'>
RangeIndex: 117 entries, 0 to 116
Data columns (total 28 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   episode_number                       117 non-null    int64
1   pitch_number                         117 non-null    int64
2   brand_name                           117 non-null    str
3   idea                                 117 non-null    str
4   deal                                 117 non-null    int64
5   pitcher_ask_amount                  117 non-null    float64
6   ask_equity                          117 non-null    float64
7   ask_valuation                       117 non-null    float64
8   deal_amount                         117 non-null    float64
9   deal_equity                         117 non-null    float64
10  deal_valuation                      117 non-null    float64
11  ashneer_present                     117 non-null    int64
12  anupam_present                      117 non-null    int64
13  aman_present                        117 non-null    int64
14  namita_present                      117 non-null    int64
15  vineeta_present                     117 non-null    int64
16  peyush_present                      117 non-null    int64
17  ghazal_present                      117 non-null    int64
18  ashneer_deal                       117 non-null    int64
19  anupam_deal                        117 non-null    int64
20  aman_deal                          117 non-null    int64
21  namita_deal                        117 non-null    int64
22  vineeta_deal                       117 non-null    int64
23  peyush_deal                        117 non-null    int64
24  ghazal_deal                        117 non-null    int64
25  total_sharks_invested               117 non-null    int64
26  amount_per_shark                    117 non-null    float64
27  equity_per_shark                    117 non-null    float64
dtypes: float64(8), int64(18), str(2)
memory usage: 25.7 KB

```

```
In [13]: df_sk_n.info()
<class 'pandas.DataFrame'>
RangeIndex: 121 entries, 0 to 120
Data columns (total 25 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Sl No.                                121 non-null    int64
1   Start-Up Name                         121 non-null    str
2   No. of Pitchers                       121 non-null    int64
3   Mean Founder Age                      121 non-null    int64
4   Sex                                    121 non-null    str
5   IDEA                                  121 non-null    str
6   Sector                                121 non-null    str
7   Past valuation                        25 non-null     float64
8   Revenue YoY                          120 non-null    float64
9   Ask_value                             119 non-null    float64
10  Ask_% of Equity                       119 non-null    str
11  Deal_value                             121 non-null    float64
12  Deal_% of Equity                       121 non-null    str
13  Deal Loan                             121 non-null    int64
14  Deal                                   121 non-null    str
15  Valuation after Deal                   67 non-null     float64
16  Aman Gupta                            121 non-null    float64
17  Ashneer Grover                        121 non-null    int64
18  Vineeta Singh                         121 non-null    float64
19  Peyush Bansal                         121 non-null    float64
20  Namita Thapar                         121 non-null    float64
21  Anupam Mittal                         121 non-null    float64
22  Ghazal Alagh                          121 non-null    float64
23  Unnamed: 23                           0 non-null      float64
24  Unnamed: 24                           0 non-null      float64
dtypes: float64(13), int64(5), str(7)
memory usage: 23.8 KB
```

```

In [15]: df.isna().any()
Out[15]:
episode_number      False
pitch_number        False
brand_name          False
idea                False
deal                False
pitcher_ask_amount  False
ask_equity          False
ask_valuation       False
deal_amount         False
deal_equity         False
deal_valuation      False
ashneer_present     False
anupam_present      False
aman_present        False
namita_present      False
vineeta_present     False
peyush_present      False
ghazal_present      False
ashneer_deal        False
anupam_deal         False
aman_deal           False
namita_deal         False
vineeta_deal        False
peyush_deal         False
ghazal_deal         False
total_sharks_invested False
amount_per_shark    False
equity_per_shark    False
dtype: bool

```

```
In [16]: df.columns
Out[16]:
Index(['episode_number', 'pitch_number', 'brand_name', 'idea', 'deal',
      'pitcher_ask_amount', 'ask_equity', 'ask_valuation', 'deal_amount',
      'deal_equity', 'deal_valuation', 'ashneer_present', 'anupam_present',
      'aman_present', 'namita_present', 'vineeta_present', 'peyush_present',
      'ghazal_present', 'ashneer_deal', 'anupam_deal', 'aman_deal',
      'namita_deal', 'vineeta_deal', 'peyush_deal', 'ghazal_deal',
      'total_sharks_invested', 'amount_per_shark', 'equity_per_shark'],
      dtype='str')
```

```
In [17]: df_sk_n.columns
Out[17]:
Index(['Sl No.', 'Start-Up Name', 'No. of Pitchers', 'Mean Founder Age', 'Sex',
      'IDEA', 'Sector', 'Past valuation ', 'Revenue YoY', 'Ask_value',
      'Ask_% of Equity', 'Deal_value', 'Deal_% of Equity ', 'Deal Loan',
      'Deal', 'Valuation after Deal', 'Aman Gupta ', 'Ashneer Grover',
      'Vineeta Singh', 'Peyush Bansal', 'Namita Thapar', 'Anupam Mittal',
      'Ghazal Alagh', 'Unnamed: 23', 'Unnamed: 24'],
      dtype='str')
```

```
In [18]: df['deal'].value_counts()
Out[18]:
deal
1      65
0      52
Name: count, dtype: int64
```

```
In [19]: df[df['pitcher_ask_amount']==df['pitcher_ask_amount'].max()]
Out[19]:
   episode_number  pitch_number  ... amount_per_shark equity_per_shark
30              11           31  ...              0.0              0.0

[1 rows x 28 columns]
```

```
In [20]: df[df['deal_amount']==df['deal_amount'].max()]
Out[20]:
   episode_number  pitch_number  ... amount_per_shark equity_per_shark
50              17           51  ...              50.0              5.0

[1 rows x 28 columns]
```

```
In [21]: df[df['ask_valuation']==df['ask_valuation'].max()]
Out[21]:
   episode_number  pitch_number  ... amount_per_shark equity_per_shark
30              11             31  ...              0.0              0.0

[1 rows x 28 columns]
```

```
In [22]: df[df['ask_equity']==df['ask_equity'].min()]
Out[22]:
   episode_number  pitch_number  ... amount_per_shark equity_per_shark
6                3             7  ...              0.0              0.0

[1 rows x 28 columns]
```

```
In [23]: df[df['deal_valuation']==df['deal_valuation'].max()]
Out[23]:
   episode_number  pitch_number  ... amount_per_shark equity_per_shark
12              5             13  ...             50.0             0.75

[1 rows x 28 columns]
```

```
In [24]: def synergy(data):
...:     list = ['anupam_deal', 'aman_deal', 'namita_deal', 'vineeta_deal', 'peyush_deal', 'ghazal_deal', 'ashneer_deal']
...:     for i in list:
...:         deal = data[['amount_per_shark', 'equity_per_shark']][data[i]==1]
...:         print("\n{} deals with {}".format(len(deal), i[:5]))
...:         print(deal)
...:
In [25]: ash_grover = df[df['ashneer_deal']==1]
...: ash_grover
Out[25]:
   episode_number  pitch_number  ... amount_per_shark equity_per_shark
0                1             1  ...      25.000000      5.333333
1                1             2  ...      20.000000      25.000000
3                2             4  ...      70.000000       2.750000
15               6            16  ...      20.000000       3.000000
18               7            19  ...      50.000000       2.000000
21               8            22  ...      25.000000       1.250000
23               8            24  ...      30.000000       6.000000
29              10            30  ...      10.000000       1.000000
38              13            39  ...      25.000000       2.500000
45              16            46  ...      25.000000       1.750000
```

49	17	50	...	10.000000	5.000000
50	17	51	...	50.000000	5.000000
55	18	56	...	1.000000	1.000000
58	19	59	...	15.000000	2.500000
63	20	64	...	20.000000	2.000000
64	21	65	...	33.333333	5.000000
67	22	68	...	20.000000	1.333333
108	33	109	...	20.000000	3.333333
109	33	110	...	5.000000	5.000000
110	34	111	...	10.000000	5.000000
114	35	115	...	10.000000	7.500000

[21 rows x 28 columns]

```
In [26]: ash_grover['amount_per_shark'].sum()
```

```
Out[26]: np.float64(494.33333333)
```

```
In [31]: df = df.dropna(subset=['total_sharks_invested'])
```

```
In [32]: df
```

```
Out[32]:
```

	episode_number	pitch_number	...	amount_per_shark	equity_per_shark
0	1	1	...	25.0	5.333333
1	1	2	...	20.0	25.000000
2	1	3	...	12.5	15.000000
3	2	4	...	70.0	2.750000
4	2	5	...	0.0	0.000000
..	...	...	...	...	...
112	34	113	...	0.0	0.000000
113	34	114	...	0.0	0.000000
114	35	115	...	10.0	7.500000
115	35	116	...	0.0	0.000000
116	35	117	...	0.0	0.000000

[117 rows x 28 columns]

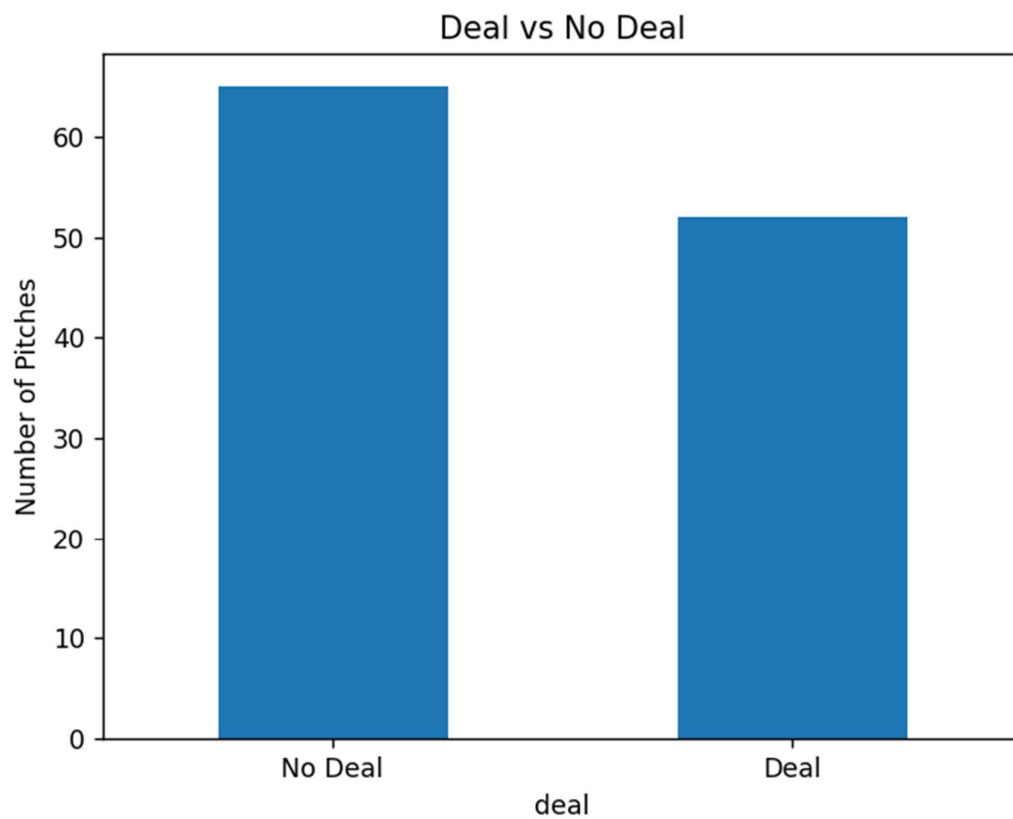
```
In [33]: deal_counts = df['deal'].value_counts()
...:
...: deal_counts
```

```
Out[33]:
```

```
deal
1    65
0    52
Name: count, dtype: int64
```

```
75 deal_counts.plot(kind='bar')
76 plt.title("Deal vs No Deal")
77 plt.xticks(ticks=[0,1], labels=["No Deal", "Deal"], rotation=0)
78 plt.ylabel("Number of Pitches")
79 plt.show()
80
```

```
In [34]: success_rate = (df['deal'].sum() / len(df)) * 100
...: print("Deal Success Rate: {:.2f}%".format(success_rate))
Deal Success Rate: 55.56%
```

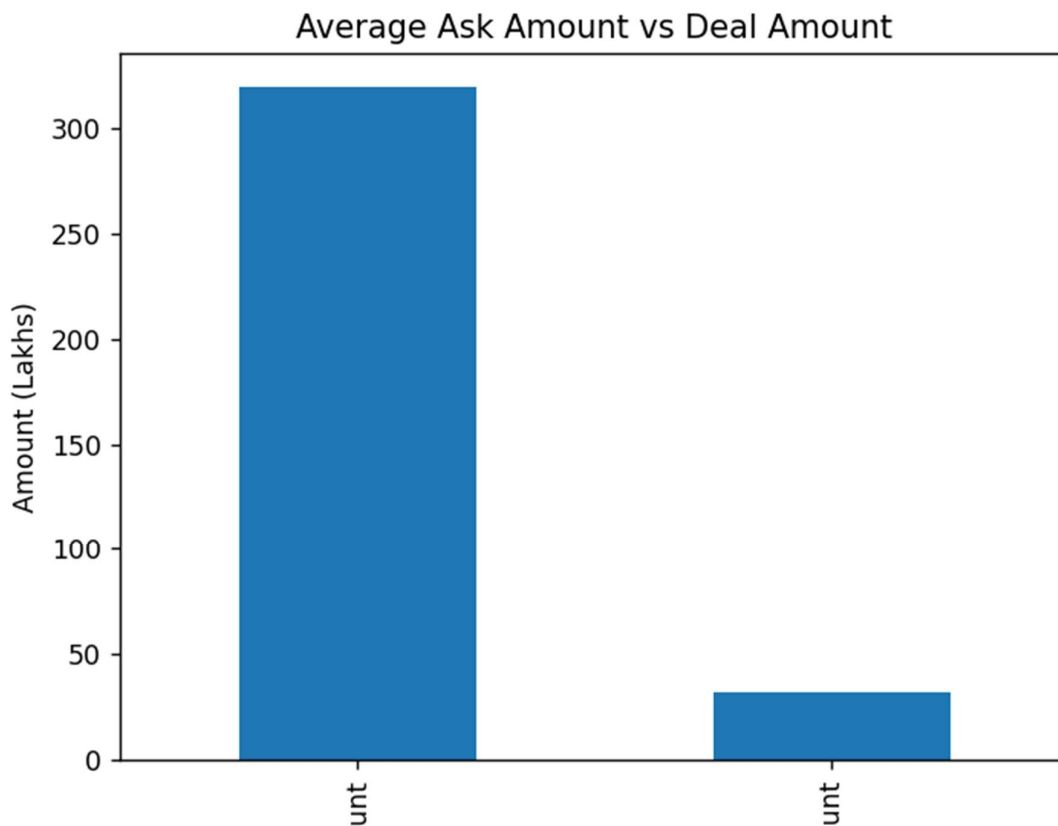


```

In [35]: df[['pitcher_ask_amount', 'deal_amount']].mean()
Out[35]:
pitcher_ask_amount    319.854709
deal_amount           31.982915
dtype: float64
In [36]: comparison = df[['pitcher_ask_amount', 'deal_amount']].mean()
In [37]: comparison
Out[37]:
pitcher_ask_amount    319.854709
deal_amount           31.982915
dtype: float64

comparison.plot(kind='bar')
plt.title("Average Ask Amount vs Deal Amount")
plt.ylabel("Amount (Lakhs)")
plt.show()

```



```

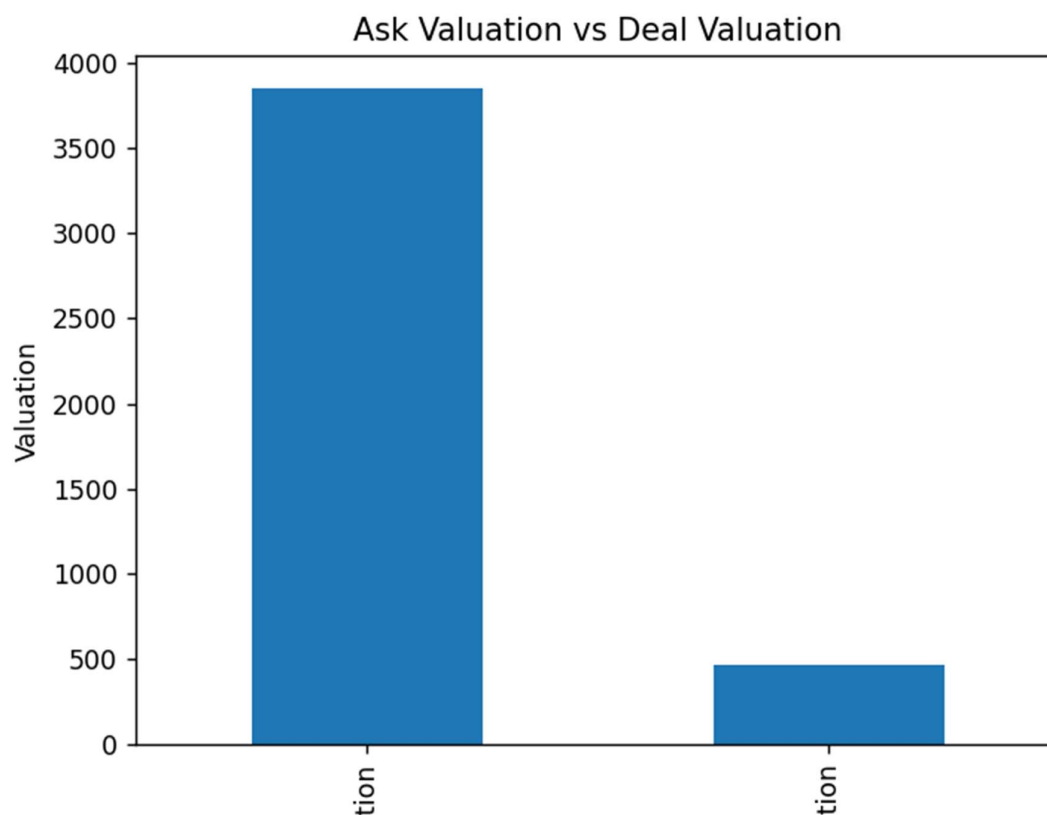
In [38]: df[['ask_valuation', 'deal_valuation']].mean()
Out[38]:
ask_valuation    3852.462479
deal_valuation    467.104872
dtype: float64
In [39]: valuation_comparison = df[['ask_valuation', 'deal_valuation']].mean()
In [40]: valuation_comparison
Out[40]:
ask_valuation    3852.462479
deal_valuation    467.104872
dtype: float64

```

```

##Visualization
valuation_comparison.plot(kind='bar')
plt.title("Ask Valuation vs Deal Valuation")
plt.ylabel("Valuation")
plt.show()

```

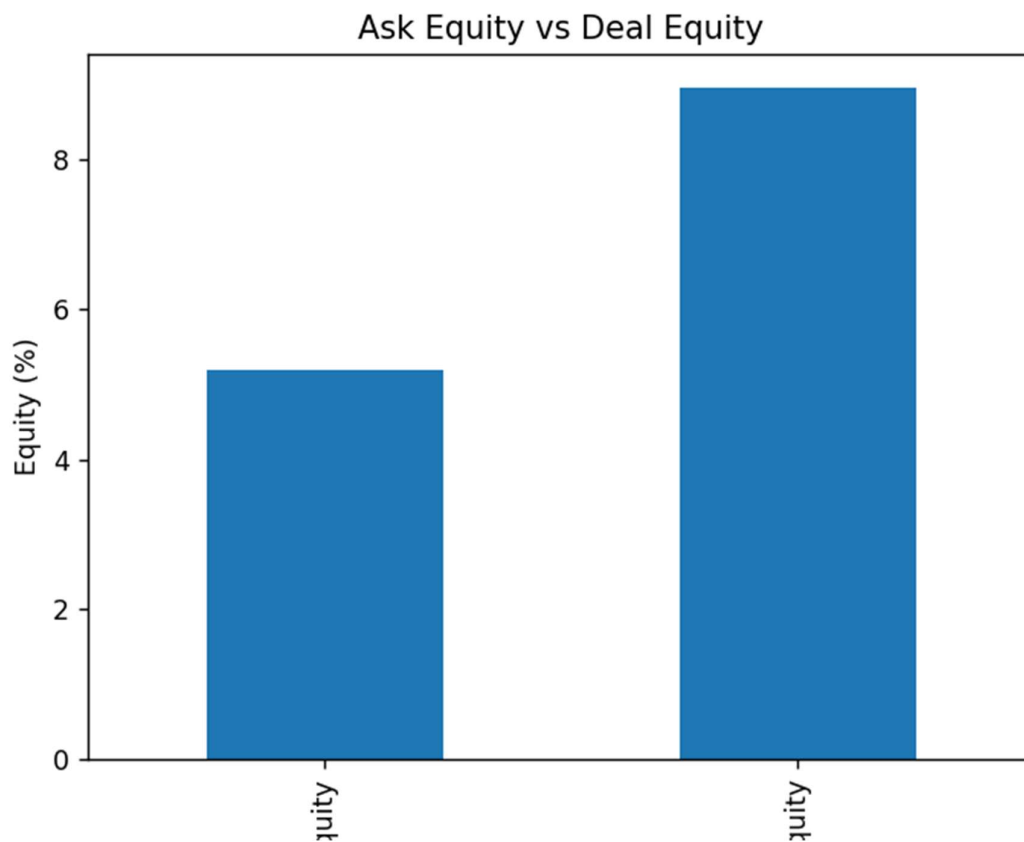


```

In [41]: df[['ask_equity', 'deal_equity']].mean()
Out[41]:
ask_equity    5.188034
deal_equity    8.963504
dtype: float64
In [42]: equity_comparison = df[['ask_equity', 'deal_equity']].mean()
...: equity_comparison
Out[42]:
ask_equity    5.188034
deal_equity    8.963504
dtype: float64

##Visualization
equity_comparison.plot(kind='bar')
plt.title("Ask Equity vs Deal Equity")
plt.ylabel("Equity (%)")
plt.show()

```



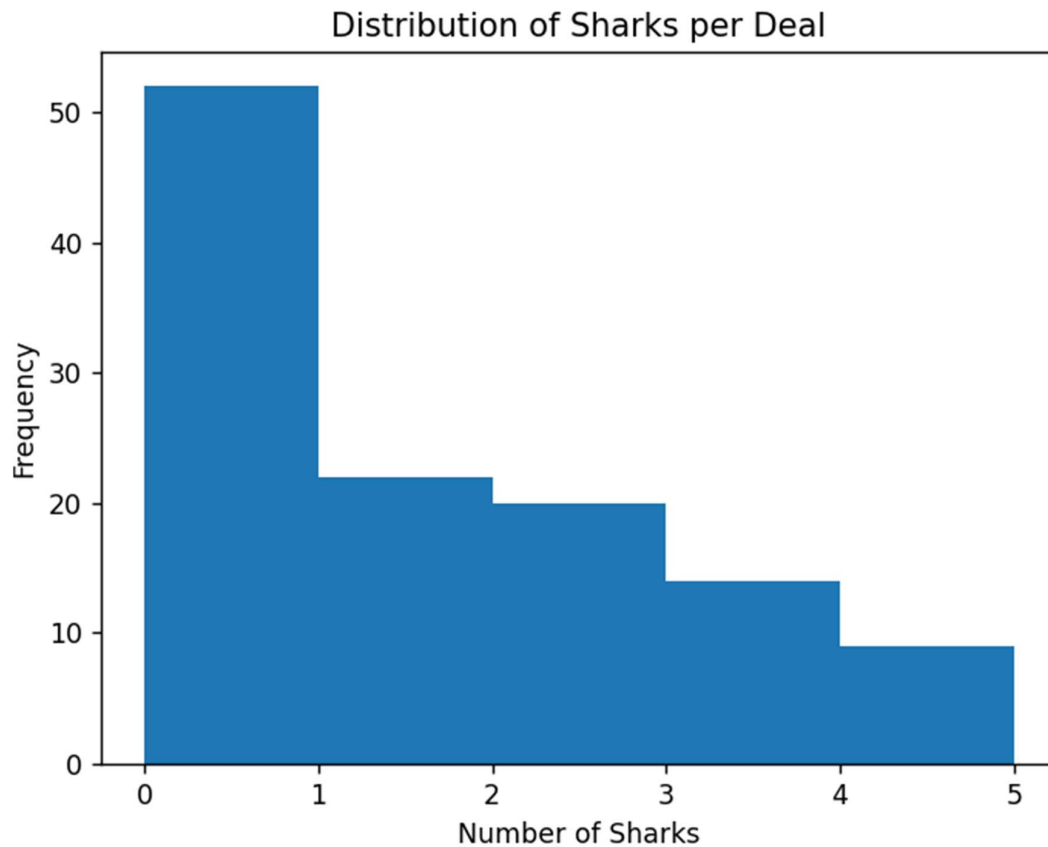
```
In [43]: df['total_sharks_invested'].describe()
```

```
Out[43]:
```

```
count    117.000000  
mean      1.230769  
std       1.410457  
min       0.000000  
25%      0.000000  
50%      1.000000  
75%      2.000000  
max       5.000000
```

```
Name: total_sharks_invested, dtype: float64
```

```
##Visualization  
df['total_sharks_invested'].plot(kind='hist', bins=5)  
plt.title("Distribution of Sharks per Deal")  
plt.xlabel("Number of Sharks")  
plt.show()
```

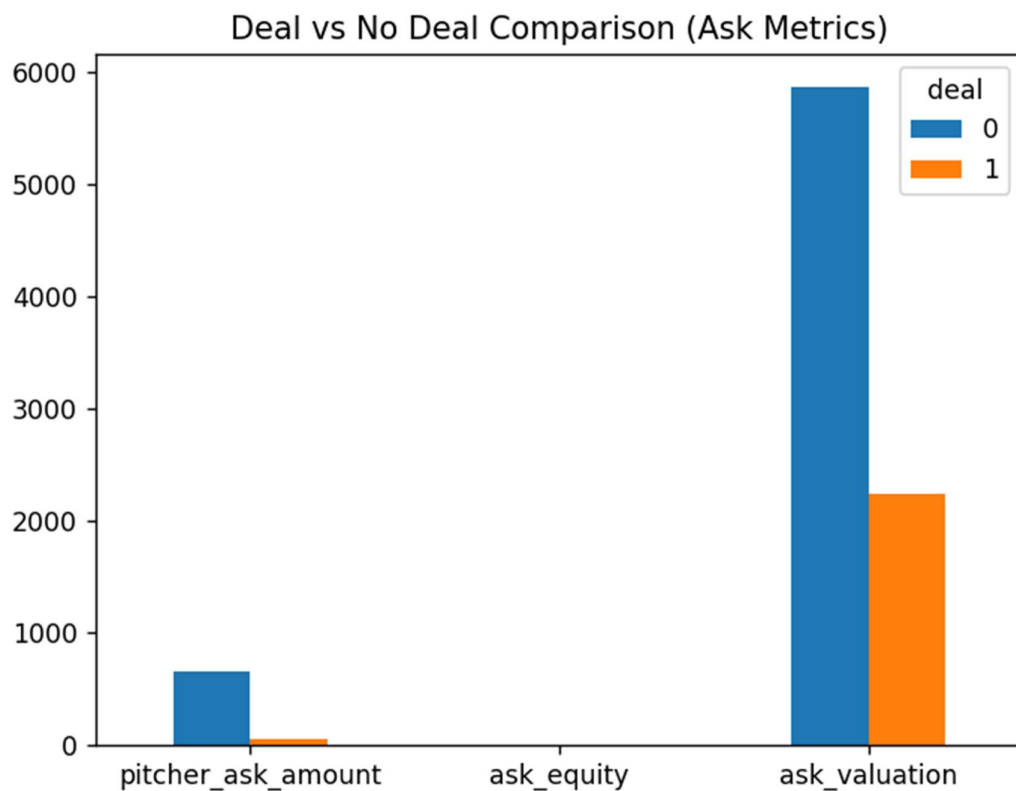


```
In [47]: deal_comparison = df.groupby('deal')[[
...:     'pitcher_ask_amount',
...:     'ask_equity',
...:     'ask_valuation'
...: ]].mean()
...: deal_comparison
```

Out[47]:

	pitcher_ask_amount	ask_equity	ask_valuation
deal			
0	649.326923	5.519231	5867.399231
1	56.276939	4.923077	2240.513077

```
##Visualization
deal_comparison.T.plot(kind='bar')
plt.title("Deal vs No Deal Comparison (Ask Metrics)")
plt.xticks(rotation=0)
plt.show()
```



```

In [48]: df['valuation_drop_%'] = (
...:     (df['ask_valuation'] - df['deal_valuation'])
...:     / df['ask_valuation']
...: ) * 100
In [49]: df[df['deal'] == 1]['valuation_drop_%'].describe()
Out[49]:
count      65.000000
mean       51.899043
std        77.796266
min       -483.333333
25%        50.000000
50%        66.667000
75%        76.190667
max        100.000000
Name: valuation_drop_%, dtype: float64

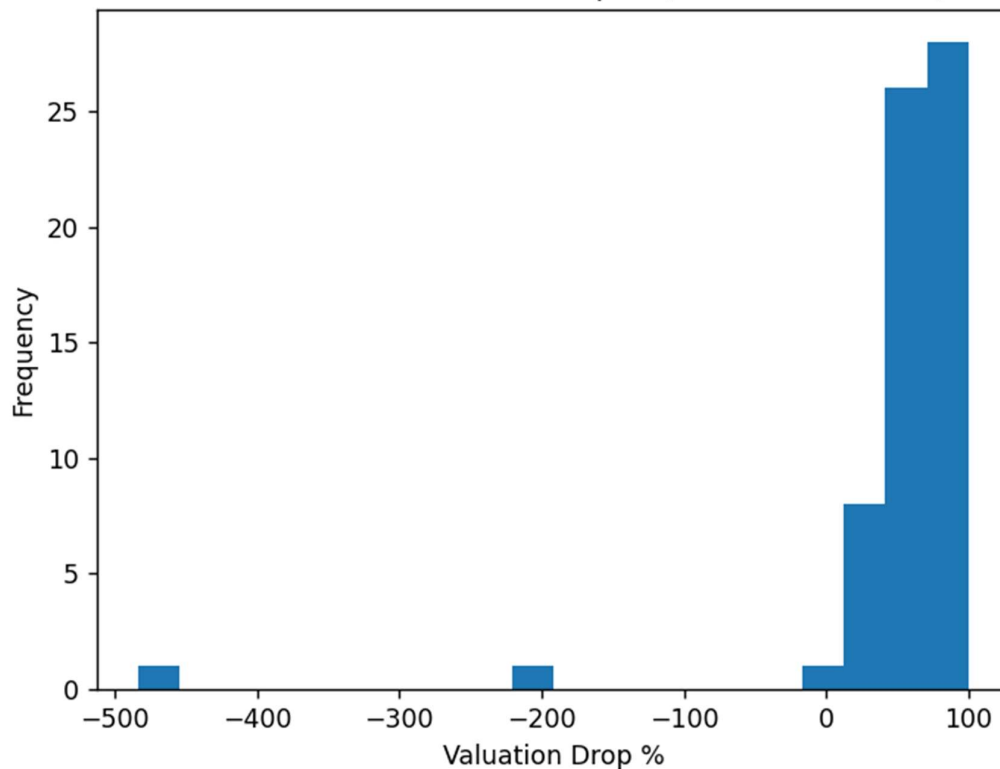
```

```

##Visualization
df[df['deal'] == 1]['valuation_drop_%'].plot(kind='hist', bins=20)
plt.title("Distribution of Valuation Drop % (Successful Deals)")
plt.xlabel("Valuation Drop %")
plt.show()

```

Distribution of Valuation Drop % (Successful Deals)



```

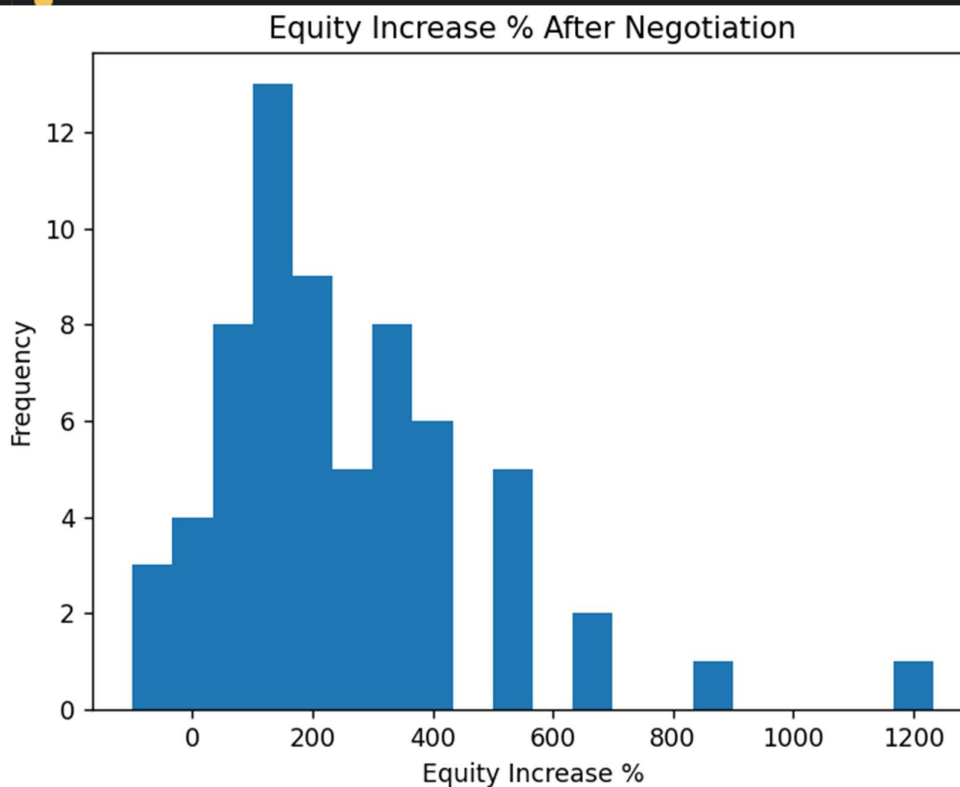
In [50]: df['equity_increase_%'] = (
...:     (df['deal_equity'] - df['ask_equity'])
...:     / df['ask_equity']
...: ) * 100
In [51]: df[df['deal'] == 1]['equity_increase_%'].describe()
Out[51]:
count      65.000000
mean       241.815385
std        228.938201
min        -100.000000
25%        100.000000
50%        200.000000
75%        337.500000
max        1233.333333
Name: equity_increase_%, dtype: float64

```

```

##Visualization
df[df['deal'] == 1]['equity_increase_%'].plot(kind='hist', bins=20)
plt.title("Equity Increase % After Negotiation")
plt.xlabel("Equity Increase %")
plt.show()

```



```
In [52]: shark_vs_amount = df[df['deal'] == 1].groupby(  
...:     'total_sharks_invested'  
...: )['deal_amount'].mean()  
...:  
...: shark_vs_amount
```

Out[52]:

total_sharks_invested

0 0.000000

1 51.409091

2 53.736842

3 64.285718

4 48.000202

5 87.500000

Name: deal_amount, dtype: float64

```
##Visualization
```

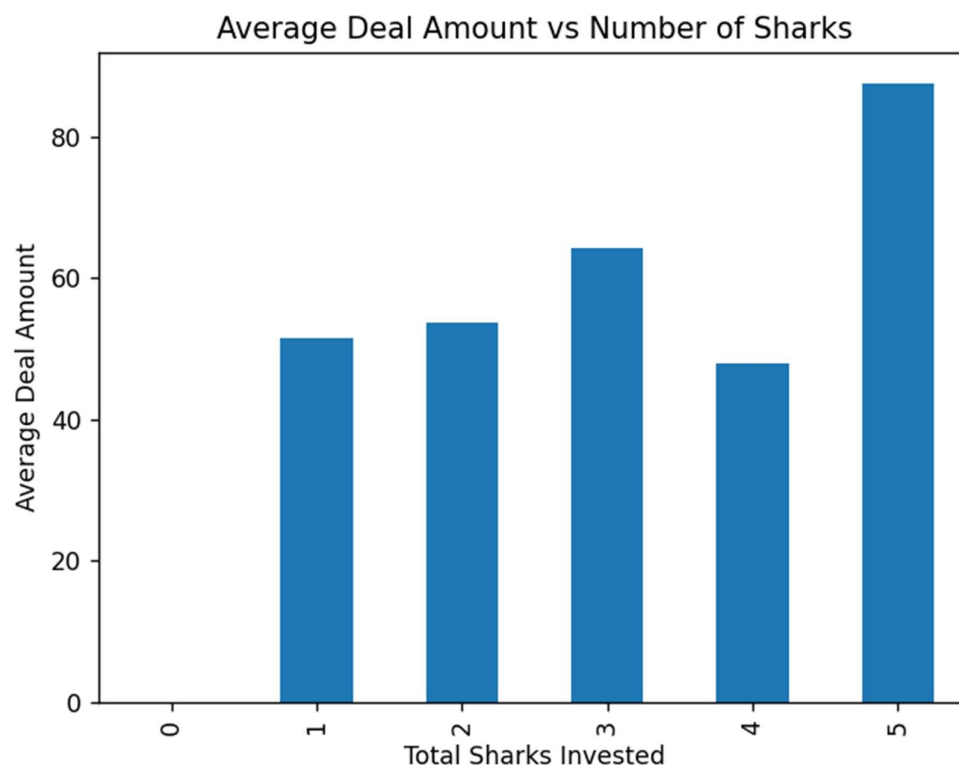
```
shark_vs_amount.plot(kind='bar')
```

```
plt.title("Average Deal Amount vs Number of Sharks")
```

```
plt.xlabel("Total Sharks Invested")
```

```
plt.ylabel("Average Deal Amount")
```

```
plt.show()
```

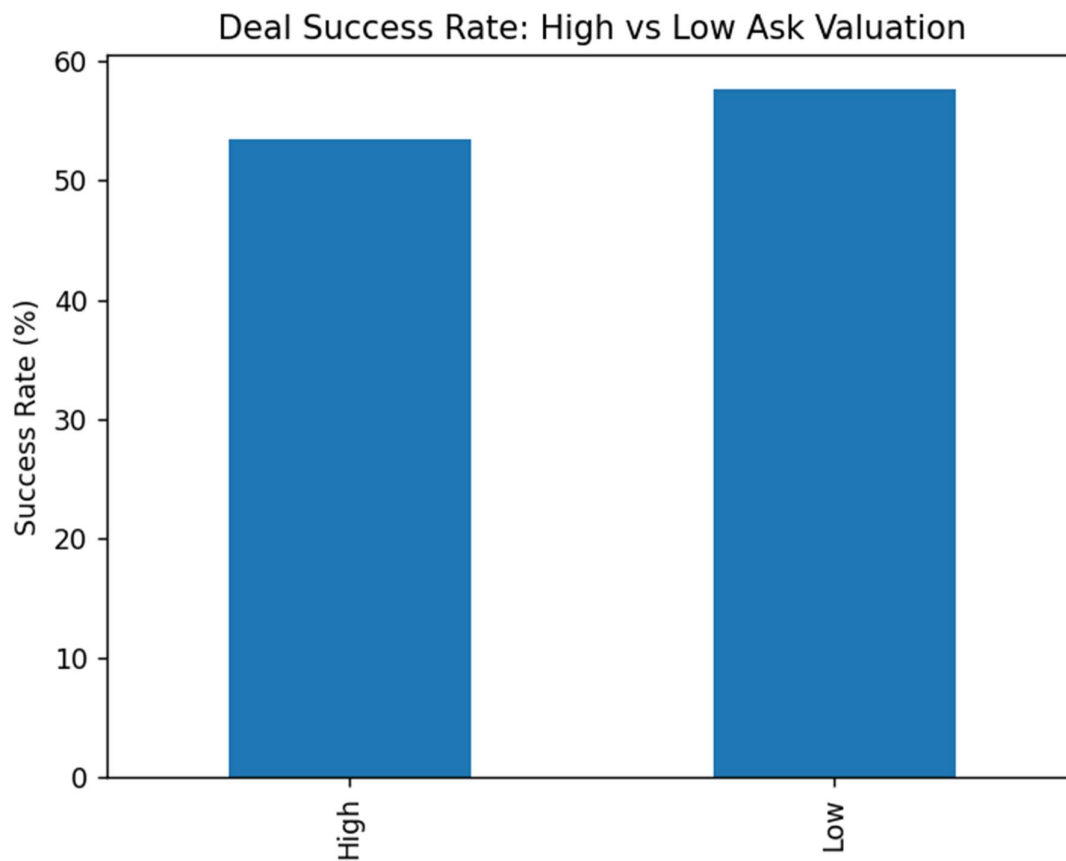


```

In [53]: median_val = df['ask_valuation'].median()
In [54]: df['valuation_category'] = df['ask_valuation'].apply(
...:     lambda x: 'High' if x > median_val else 'Low'
...: )
In [55]: valuation_success = df.groupby('valuation_category')['deal'].mean() * 100
In [56]: valuation_success
Out[56]:
valuation_category
High      53.448276
Low       57.627119
Name: deal, dtype: float64

##Visualization
valuation_success.plot(kind='bar')
plt.title("Deal Success Rate: High vs Low Ask Valuation")
plt.ylabel("Success Rate (%)")
plt.show()

```

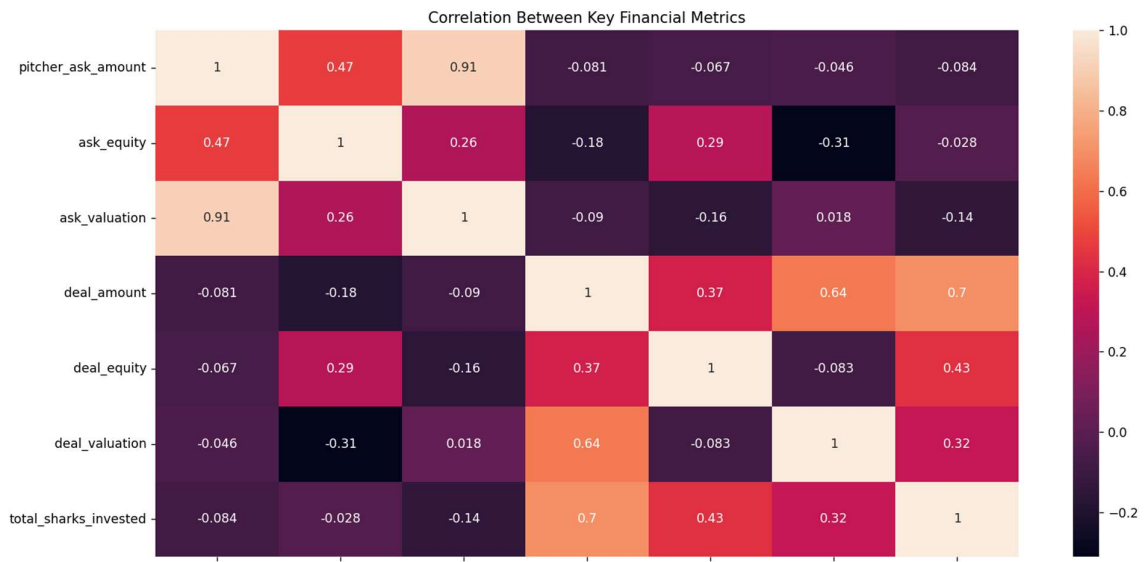


```

important_cols = [
    'pitcher_ask_amount',
    'ask_equity',
    'ask_valuation',
    'deal_amount',
    'deal_equity',
    'deal_valuation',
    'total_sharks_invested'
]

##Visualization
sns.heatmap(df[important_cols].corr(), annot=True)
plt.title("Correlation Between Key Financial Metrics")
plt.show()

```



```

In [59]: df[df['deal_amount'] > df['deal_amount'].quantile(0.95)][
...:      ['deal_amount', 'total_sharks_invested', 'deal_equity']
...: ]

```

```

Out[59]:
   deal_amount  total_sharks_invested  deal_equity
36        105.0                      3          3.0
50        150.0                      3         15.0

```